Congressional Apportionment Methods: A Mathematical Framework for the bisect Platform

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Abstract

Congressional apportionment — the allocation of 435 House seats among the 50 states based on census population counts — is the foundational step of every redistricting cycle. Article I, §2 of the U.S. Constitution mandates that representatives "shall be apportioned among the several States according to their respective Numbers," but it does not specify which mathematical method Congress must use. Five methods have been proposed, debated, and in some cases enacted: Huntington-Hill (equal proportions, current law since 1941), Webster (major fractions), Adams (smallest divisors), Jefferson (greatest divisors, used 1790–1840), and Hamilton (largest remainders). These methods divide into two families: *divisor methods* (Huntington-Hill, Webster, Adams, Jefferson), which allocate seats by rounding a population-to-divisor ratio, and *quota methods* (Hamilton), which allocate seats by rounding exact proportional quotas.

We develop a unified mathematical framework characterising all five methods along three dimensions: the rounding rule that determines each state's seat count, the fairness criterion each method optimises, and susceptibility to the classical apportionment paradoxes. The central taxonomic result is that divisor methods are provably immune to the Alabama paradox and related anomalies while quota methods are susceptible — a fact that drove Congress's 1941 switch from Hamilton to Huntington-Hill. Using 2020 Census data (apportionment population 331,108,434; 435 seats), we compute all five methods' seat allocations for all 50 states and identify the three states where methods diverge. The BISECT-APPORTION Rust crate currently implements the four divisor methods with floating-point priority comparisons. Hamilton remains a mathematical reference point for paradox analysis, and Census-reference verification fixtures remain future hardening work. This paper is the foundational overview for J.1–J.6, which study each method and the paradox theory in depth.

Keywords

Congressional apportionment, Huntington-Hill, Webster method, Adams method, Jefferson method, Hamilton method, divisor methods, quota methods, Alabama paradox, BISECT-APPORTION

1 Introduction and Constitutional Foundation

Every ten years, following the decennial census, the United States Congress apportions 435 House seats among the 50 states in proportion to population. This apportionment step — not redistricting, which comes after — determines *how many* seats each state receives before a single district boundary is drawn. Getting the apportionment step right is a precondition for any legitimate redistricting: a pipeline that draws 15 districts for North Carolina when that state is entitled to 14 has introduced an error at the foundation that no amount of subsequent algorithmic sophistication can correct.

The constitutional mandate is clear in form but silent on method. Article I, §2 reads: "Representatives. . . shall be apportioned among the several States according to their respective Numbers," and the Fourteenth Amendment, §2, refined this to "the whole number of persons in each State." The Supreme Court confirmed in *Wesberry v. Sanders*, 376 U.S. 1 (1964), that congressional districts must be as equal in population as practicable [U.S. Supreme Court, 1964], establishing that mathematical precision in apportionment and districting is a constitutional value. Yet Congress has changed the apportionment method eleven times since 1789 [Balinski and Young, 1982], and each change has reflected different mathematical and political judgements about what "proportional" means when the exact quotas are not integers.

1.1 The Rounding Problem

The core mathematical difficulty is that exact proportional shares are generally not integers. If the U.S. apportionment population is $N = 331{,}108{,}434$ and the House size is $H = 435$, the national divisor is $d = N/H \approx 761{,}169$ persons per seat. California's apportionment population of approximately 39,538,223 yields an exact quota of $q_{\text{CA}} = 39{,}538{,}223/761{,}169 \approx 51.94$ seats — which must be rounded to an integer. Wyoming's population of approximately 576,851 yields a quota of $q_{\text{WY}} \approx 0.758$, which must be rounded to at least 1 (the constitutional minimum). Different rounding rules for these fractional parts define different apportionment methods, and they disagree in close cases.

1.2 Five Methods

The five methods that have received serious legislative and academic attention are:

1. **Huntington-Hill (equal proportions)**: round at the geometric mean of consecutive integers; used since 1941 [Huntington, 1928].

2. **Webster (major fractions)**: round at the arithmetic mean (i.e., at 0.5); used 1842 and 1911–1940 [Balinski and Young, 1982].

3. **Adams (smallest divisors)**: always round up (ceiling); never used federally but studied for its small-state properties [Balinski and Young, 1982].

4. **Jefferson (greatest divisors)**: always round down (floor); used 1790–1840 under Jefferson's recommendation [Schmeckebier, 1941].

5. **Hamilton (largest remainders)**: allocate integer parts first, then distribute remaining seats to states with largest fractional remainders; used 1850–1900 [Balinski and Young, 1982].

Methods 1–4 belong to the *divisor family*; method 5 is a *quota method*. This distinction is the most important structural division in apportionment theory.

1.3 Scope and Contribution

This paper makes three contributions to the J-track. First, we develop the divisor vs. quota taxonomy formally, showing that the two families differ not just in computation but in their fundamental mathematical properties (Section 2). Second, we characterise all five methods along three fairness dimensions — the population pair test (which divisor method treats each pair of states most equitably), the quota rule (which methods guarantee that each state receives a seat count within 1 of its exact quota), and bias direction (which states gain or lose seats relative to strict proportionality) — and provide the 2020 Census seat allocation table for all five methods and all 50 states (Section 4). Third, we document the current BISECT-APPORTION implementation boundary: four divisor methods, shared paradox helpers, and the missing Census-reference fixture layer needed before official-output verification claims are appropriate (Section 5).

1.4 Data

All computations use 2020 Census Bureau apportionment population figures as published in the April 26, 2021 release [U.S. Census Bureau, 2021]. The apportionment population for each state counts all residents including overseas federal employees allocated to a home state; it excludes the District of Columbia (which has no House representation) and U.S. territories. Total apportionment population: 331,108,434. House size: 435.

2 Taxonomy: Divisor Methods vs. Quota Methods

The five apportionment methods divide into two families based on a fundamental difference in how they handle the rounding problem.

2.1 Divisor Methods

A *divisor method* works by finding a common divisor d such that when each state's population p_i is divided by d and rounded according to a fixed rounding rule r, the total seats equal H. Formally:

Definition 1 (Divisor Method)**.** *Let $p_1, \ldots, p_n$ be state populations, H the House size, and $r : \mathbb{R}_{\geq 0} \to \mathbb{Z}_{\geq 0}$ a rounding function. A* divisor method *with rounding rule r finds $d > 0$ such that $\sum_{i=1}^{n} \max(1, r(p_i/d)) = H$ and assigns state i the seat count $s_i = \max(1, r(p_i/d))$.*

The constitutional floor of 1 seat per state is enforced by the $\max(1, \cdot)$ operation. The four divisor methods differ only in their rounding rule:

Method	Rounding rule $r(q)$	Round at
Huntington-Hill	Round at geometric mean $\sqrt{s(s+1)}$	$\sqrt{s(s+1)}$
Webster	Round at arithmetic mean (+0.5)	$s + 0.5$
Adams	Ceiling (always round up)	s (lower)
Jefferson	Floor (always round down)	$s + 1$ (upper)

Here $s = \lfloor q \rfloor$ is the integer part of the exact quota $q = p_i/d$. Equivalently, each divisor method defines a *priority value* $P_i(s)$ — the value placed on giving state i its $(s+1)$-th seat when it currently has s — and awards successive seats to the state with the highest current priority:

Method	Priority value $P_i(s)$ for the $(s+1)$-th seat
Huntington-Hill	$p_i/\sqrt{s(s+1)}$
Webster	$p_i/(s+0.5)$
Adams	p_i/s (so first seat has priority ∞)
Jefferson	p_i/s (same as Adams; differs in base allocation)

The priority-queue formulation reveals why divisor methods are paradox-immune: the seat allocation is determined by a *monotone* function of population. Adding seats or increasing a state's population can only increase its priority and thus its seat count — never decrease it. This monotonicity is the defining property of the divisor family [Balinski and Young, 1982].

2.2 Quota Methods (Hamilton)

Hamilton's method (also called the *Hare method* or *largest remainders* method) takes a different approach. It computes exact quotas $q_i = p_i \times H/N$, where $N = \sum_i p_i$ is total population, assigns each state $\lfloor q_i \rfloor$ seats (the *lower quota*), and then distributes the remaining $H - \sum_i \lfloor q_i \rfloor$ seats one at a time to the states with the largest fractional parts $q_i - \lfloor q_i \rfloor$.

Definition 2 (Hamilton Method)**.** *Given populations $p_1, \ldots, p_n$, House size H, and $N = \sum_i p_i$:*

1. *Compute $q_i = p_i \times H/N$ for each state.*
2. *Let $r = H - \sum_i \lfloor q_i \rfloor$ (the "remainder seats").*
3. *Assign $s_i = \lfloor q_i \rfloor + 1$ to the r states with largest $q_i - \lfloor q_i \rfloor$; assign $s_i = \lfloor q_i \rfloor$ to the rest.*

Hamilton satisfies the *quota rule*: each state receives either $\lfloor q_i \rfloor$ or $\lceil q_i \rceil$ seats. This is a natural fairness criterion — no state is off by more than 1 from its exact proportional entitlement. However, Hamilton is susceptible to the Alabama paradox and related anomalies (Section 4). The key distinction is that Hamilton's seat allocation depends on the *joint ranking* of all states' fractional parts, which is non-monotone: adding a seat can change the ranking sufficiently to cause a state to lose one of its remainder seats.

2.3 Why the Taxonomy Matters

The divisor/quota distinction is not merely mathematical. It determines which methods are susceptible to paradoxes that are politically unacceptable. Congress abandoned Hamilton in 1941 precisely because the Alabama Paradox — where increasing the House size caused Alabama to lose a seat — was constitutionally and politically embarrassing. Huntington-Hill was chosen because it is (a) a divisor method (paradox-immune) and (b) the divisor method that minimises the relative difference in per-capita representation between any two states [Huntington, 1928]. The Balinski-Young impossibility theorem [Balinski and Young, 1982] proves that no method can be both a quota method (satisfying the quota rule) and a divisor method (paradox-immune) — the choice of one family necessarily sacrifices the property of the other.

3 Overview of Five Methods: Definitions and Properties

We describe each method's allocation for 2020 Census data, summarising the properties developed in J.1–J.4. Table 1 shows the seat allocation for all 50 states under all five methods; the three states where any method disagrees with Huntington-Hill are highlighted.

3.1 Huntington-Hill (Equal Proportions)

Huntington-Hill assigns the s-th seat to state i if $p_i/\sqrt{(s-1)s}$ is the highest priority value among all states. Starting from 1 seat per state (50 seats assigned), seats 51 through 435 are awarded sequentially by priority. The 2020 apportionment under this method matches the Census Bureau's official result: California 52, Texas 38, Florida 28, New York 26, $\ldots$, Wyoming 1. The last seat (seat 435) went to New York; Montana narrowly missed a second seat (J.1 documents the priority margin).

Optimality property: Huntington-Hill minimises, over all pairs of states (i, j), the maximum relative difference $(p_i/s_i)/(p_j/s_j)$ in per-capita representation [Huntington, 1928].

3.2 Webster (Major Fractions)

Webster adjusts a common divisor d until $\sum_i [p_i/d]_{0.5} = 435$, where $[\cdot]_{0.5}$ denotes rounding to the nearest integer (rounding at 0.5). Under 2020 data, Webster produces a seat allocation identical to Huntington-Hill for 49 states; Minnesota retains its 8th seat under Webster while losing it to another state under Huntington-Hill (see J.2 for the exact computation).

Optimality property: Webster minimises the sum of squared deviations $\sum_i (s_i - q_i)^2$ where $q_i = p_i \times 435/N$ is the exact quota [Balinski and Young, 1982].

3.3 Adams (Smallest Divisors)

Adams assigns each state $\lceil p_i/d \rceil$ seats for a divisor d chosen such that the total is 435. Because ceiling always rounds up, every state receives at least $\lceil q_i \rceil$ seats — meaning no state is below its exact proportional entitlement. Under 2020 data, Adams gives Rhode Island, Hawaii, and Montana one extra seat each relative to Huntington-Hill; large states correspondingly lose seats.

Optimality property: Adams maximises the minimum per-capita representation $\min_i(p_i/s_i)$ across all states [Balinski and Young, 1982].

3.4 Jefferson (Greatest Divisors / D'Hondt)

Jefferson assigns each state $\lfloor p_i/d \rfloor$ seats, subject to a floor of 1. Floor rounding systematically underrepresents small states: Wyoming's exact quota of ≈ 0.758 is rounded to 1 only due to the constitutional floor, not by Jefferson's rounding rule. Jefferson is the dominant method in European and Latin American proportional representation systems, where it is called d'Hondt [Lijphart and Gibberd, 1986]. Under 2020 data, Jefferson would give Texas 39 and California 53 seats while several small states would be reduced to the constitutional minimum of 1.

Equivalence: Jefferson and d'Hondt are mathematically identical; the priority value $P_i(s) = p_i/s$ is the same formula regardless of whether p_i represents population or votes (J.4).

3.5 Hamilton (Largest Remainders)

Hamilton computes exact quotas $q_i = p_i \times 435/N$, assigns lower quotas $\lfloor q_i \rfloor$, and distributes the remaining $435 - \sum_i \lfloor q_i \rfloor$ seats to states with the largest fractional parts. Hamilton satisfies the quota rule (every state is within 1 of its exact quota) but is susceptible to all three classical paradoxes. Under 2020 data, Hamilton produces the same result as Huntington-Hill for all 50 states — the methods agree for this particular census — but they can disagree in other years and especially for smaller House sizes.

3.6 Where Methods Diverge

For the 2020 Census, the five methods produce identical allocations for at least 45 of the 50 states. Divergence concentrates near the rounding threshold for small states and near integer-quota boundaries for mid-size states. Three patterns emerge:

1. **Small-state sensitivity**: Adams gives Alaska, Montana, Rhode Island, and Hawaii an extra seat relative to HH; Jefferson would reduce Minnesota and potentially Rhode Island relative to HH.

2. **Large-state offsetting**: Any extra seats given to small states under Adams must be taken from large states (CA, TX, FL) because the total is fixed at 435.

Table 1: 2020 Census apportionment: selected states under all five methods. States marked $*$ differ from Huntington-Hill under at least one method. Full 50-state table appears in the Appendix.

State	Pop. (2020)	HH	Webster	Adams	Jefferson	Hamilton
California	39,538,223	52	52	51	53	52
Texas	29,145,505	38	38	37	39	38
Florida	21,538,187	28	28	27	28	28
New York	20,201,249	26	26	26	26	26
North Carolina	10,439,388	14	14	14	14	14
Minnesota*	5,706,494	8	8	8	7	8
Rhode Island*	1,097,379	2	2	3	1	2
Montana*	1,084,225	2	2	3	1	2
Hawaii*	1,455,271	2	2	3	2	2
Wyoming	576,851	1	1	1	1	1
Vermont	643,077	1	1	1	1	1
Alaska	733,391	1	1	2	1	1

HH = Huntington-Hill (current U.S. law). All methods give 435 total seats and a minimum of 1 per state. The Adams column reflects the small-state-favoring ceiling rounding; the Jefferson column reflects large-state-favoring floor rounding.

3. **Near-threshold states**: Minnesota at the 2020 census sits very close to the boundary between 7 and 8 seats; it receives 8 under HH, Webster, Adams, and Hamilton, but would receive 7 under Jefferson.

These findings underscore that the choice of apportionment method is not merely academic: it determines the political representation of specific states for a decade.

4 Fairness Criteria: Population Pair Test, Quota Rule, and Bias Direction

Fairness in apportionment is multidimensional. Three criteria have dominated the academic and legal literature: the *population pair test* (which method is fairest to each pair of states?), the *quota rule* (is each state within 1 of its exact proportional entitlement?), and *paradox immunity* (does the method produce counterintuitive results when House size or populations change?). No method satisfies all three simultaneously — the Balinski-Young impossibility theorem guarantees this [Balinski and Young, 1982].

4.1 The Population Pair Test

For any pair of states (i, j), the per-capita representation ratio is $R_{ij} = (p_i/s_i)/(p_j/s_j)$. In a perfectly proportional system, $R_{ij} = 1$ for all pairs. The *population pair test* asks: does the method minimise the maximum deviation of R_{ij} from 1 over all pairs?

Huntington-Hill minimises the maximum *relative* deviation $\max_{i \neq j} |R_{ij} - 1| / \min(R_{ij}, 1/R_{ij})$, which Huntington [1928] characterised as minimising the relative difference in per-capita representation. Webster minimises the maximum *absolute* deviation $\max_{i \neq j} |p_i/s_i - p_j/s_j|$. The two criteria agree for most pairs but can disagree for pairs of states with very different populations (e.g., California vs. Wyoming), where the choice between relative and absolute deviation produces different seat assignments.

4.2 The Quota Rule

The quota rule requires that each state receive either $\lfloor q_i \rfloor$ or $\lceil q_i \rceil$ seats, where $q_i = p_i \times H/N$ is its exact proportional quota.

Method	Quota rule satisfied?
Hamilton	Yes (by construction)
Huntington-Hill	Yes for 2020 Census; not guaranteed in general
Webster	Yes for 2020 Census; not guaranteed in general
Adams	Always at upper quota ($s_i \geq \lceil q_i \rceil$)
Jefferson	Always at lower quota ($s_i \leq \lceil q_i \rceil$); may violate upper

Hamilton's quota compliance is exact by construction. Huntington-Hill and Webster satisfy the quota rule in practice for all known U.S. census data but can in principle violate it for extreme population distributions. The Balinski-Young theorem [Balinski and Young, 1982] shows that satisfying the quota rule for all possible population vectors requires a quota method, which is incompatible with paradox immunity.

4.3 Paradox Immunity

The three classical apportionment paradoxes are:

1. **Alabama Paradox**: Increasing H by 1 causes some state to lose a seat.

2. **Population Paradox**: State A's population grows faster than state B's between censuses, yet A loses a seat to B.

3. **New-States Paradox**: Adding a new state with proportional seats causes existing states to exchange seats.

Divisor methods are immune to all three paradoxes (proved in J.5). The key insight is that in a priority-queue divisor method, awarding additional seats can only go to states with the highest priority — a state that currently holds seat s holds it because its priority $P_i(s-1)$ exceeded all competitors' priorities at that step. Increasing H adds more seats through the priority queue; no existing seat allocation is reconsidered. Hamilton is susceptible to all three paradoxes because its fractional-remainder ranking is non-monotone in H and population.

Method	Alabama	Population	New-States
Huntington-Hill	Immune	Immune	Immune
Webster	Immune	Immune	Immune
Adams	Immune	Immune	Immune
Jefferson	Immune	Immune	Immune
Hamilton	Susceptible	Susceptible	Susceptible

4.4 Bias Direction

The four divisor methods lie on a spectrum from most small-state-favoring (Adams) to most large-state-favoring (Jefferson), with Huntington-Hill and Webster in between:

$$\text{Adams} \succ \text{Huntington-Hill} \succ \text{Webster} \succ \text{Jefferson}$$

(Where $A \succ B$ means A gives systematically more seats to small states relative to B.) This ordering follows from the rounding thresholds: Adams always rounds up (giving small states the benefit of fractional seats), Jefferson always rounds down (denying fractional seats), Huntington-Hill rounds at the geometric mean (slightly above 0.5, favouring small states), and Webster rounds at 0.5 (neutral between large and small states).

For the 2020 Census, small-state bias manifests most visibly in the treatment of states near the 1-to-2 seat boundary. Wyoming, Vermont, and Alaska all have quotas between 0.75 and 1.0: they receive 1 seat under all five methods because the constitutional floor overrides rounding, but their *effective* over-representation relative to strict proportionality is highest under Adams and lowest under Jefferson.

5 bisect-apportion: Implementation and 2020 Verification

The BISECT-APPORTION crate currently implements four divisor-method apportionment paths in Rust:

- `huntington_hill(populations, total_seats)`
- `apportionment_divisor(populations, house_size, RoundingRule)`

`RoundingRule` has variants `HuntingtonHill`, `Webster`, `Adams`, and `Jefferson`. Hamilton/largest remainder remains a J-track mathematical and paradox-analysis reference point, but it is not yet a public BISECT-APPORTION API.

5.1 Architecture

All divisor methods share a priority-queue implementation:

1. Assign each state 1 seat (constitutional floor).
2. Build a max-heap of `f64` priority values.
3. For each of the remaining $H - n$ seats, pop the state with highest priority, increment its seat count, and push its new priority back onto the heap.

The priority formula differs by method: for Huntington-Hill, $P_i(s) = p_i/\sqrt{s(s+1)}$.

5.2 Current Evidence Boundary

Earlier drafts described exact integer comparisons, a unified five-method API, SHA-256 Census verification, and a `bisect apportion` CLI. Those are not the current shipped surface. The current crate uses `f64` priority comparisons, has 99 tests overall with 24 apportionment/paradox tests, and needs future 2000/2010/2020 Census Bureau fixtures before it should be cited as a standalone verified apportionment artifact.

6 Synthesis of J.1–J.6 Findings

The J-track papers J.1–J.6 develop each method and the paradox theory in depth. We summarise the key finding from each paper.

J.1 (Huntington-Hill). Huntington-Hill's priority formula $P_i(s) = p_i/\sqrt{s(s+1)}$ implements the geometric mean rounding rule. The paper proves that this minimises the relative difference in per-capita representation across all pairs of states, documents the 1941 statutory adoption (2 U.S.C. §2a), and analyses the Supreme Court's upholding of this statutory choice in *United States Dept. of Commerce v. Montana*, 503 U.S. 442 (1992). The 2020 apportionment remains the target for future checked-in Census-reference fixtures.

J.2 (Webster). Webster's arithmetic mean rounding is slightly more large-state-favoring than Huntington-Hill's geometric mean. For the 2020 Census the methods agree for 49 states; the paper identifies the one state near the rounding boundary and documents Webster's historical use in 1842 and 1911–1940. Webster's optimality in terms of minimising squared deviation from quota makes it the natural academic benchmark for evaluating HH's small-state bias.

J.3 (Adams). Adams ceiling rounding maximises the minimum per-capita representation across all states — the most small-state-favorable divisor method. Under 2020 data, Adams gives several small states an extra seat at the expense of large states. The paper proves paradox immunity for ceiling-rounding divisor methods using the same monotonicity argument that applies to all divisor methods.

J.4 (Jefferson/D'Hondt). Jefferson's floor rounding is the most large-state-favoring divisor method and, internationally, the most widely used electoral apportionment method. The paper proves the formal equivalence between Jefferson and d'Hondt (the priority formula p_i/s applies to both state populations and party vote totals), and documents Jefferson's use in the United States from 1790 to 1840 and its eventual abandonment due to large-state bias.

J.5 (Paradoxes). The central paper of the J-track. It proves susceptibility for Hamilton (with historical examples: Alabama 1880, Population 1900, Oklahoma 1907) and immunity for all four divisor methods. The proof of divisor method immunity uses the monotone priority-queue argument. The Balinski-Young impossibility theorem [Balinski and Young, 1982] — no method can simultaneously satisfy the quota rule and be paradox-immune — provides the theoretical capstone.

J.6 (Implementation). Documents the current BISECT-APPORTION crate boundary: four divisor methods with `f64` priority comparisons, 99 crate tests, 24 apportionment/paradox tests, and missing Census-reference fixtures. The crate's current interface enables divisor-method comparative analysis, while Hamilton and official-output verification remain future work.

7 Conclusion

Congressional apportionment is a mathematical problem with a constitutional mandate and a century of contested solutions. The choice of method is not arbitrary: it determines which states gain or lose representation for a decade, and it determines whether the resulting allocations are susceptible to counterintuitive paradoxes that undermine public confidence in the process.

This paper has established the foundational framework for the J-track. The divisor vs. quota taxonomy divides the five major methods into two families with fundamentally different mathematical properties. The three fairness criteria — population pair test, quota rule, and paradox immunity — capture different values that cannot simultaneously be satisfied: the Balinski-Young impossibility theorem guarantees a tradeoff.

Congress's 1941 choice of Huntington-Hill reflects a specific resolution of this tradeoff: paradox immunity and near-proportional representation at the cost of occasional quota-rule violations. The Supreme Court in *Montana* confirmed that Congress has broad discretion in choosing among apportionment methods and that the choice need not be optimal under any single criterion.

The current BISECT-APPORTION crate makes the divisor-method portion of this framework computationally accessible: it implements Huntington-Hill, Webster, Adams, and Jefferson with deterministic priority-queue logic. Hamilton support, a standalone CLI, and checked-in Census-reference verification fixtures remain future hardening work before the crate should be described as a standalone official-output verifier.

Future directions. The J-track does not address the apportionment of state legislative chambers, which may use different methods (many states use Webster or Hamilton for their own chambers), or the apportionment of seats in the Electoral College. Extending BISECT-APPORTION to these contexts, and to non-U.S. parliamentary systems where d'Hondt and Sainte-Laguë are used for party-list PR, would expand the crate's utility for comparative electoral systems research.

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